



Degree	University/Institute	Year	CGPA/ Marks(%)
M.tech Sustainable Engineering - TA	IIT Hyderabad	2026	8.92
B.Tech Electronics and Communication Engineering	Madan Mohan Malaviya University of Technology, Gorakhpur	2024	8.16
XII (ISC)	SKD ACADEMY	2019	88.00%
X (ICSE)	SKD ACADEMY	2017	87.83%

SKILLS

Technical Skills : Data Science and Analysis (Pandas, NumPy, Matplotlib, Scikit-Learn), Optimization (Linear & Stochastic Programming), Supply chain design and analysis, Circular Economy, SWOT Analysis.

Programming Language : Java, Python, R, Data Structure and Algorithm.

Programming Language : Java, R, Python

Web Development : HTML, CSS, JavaScript

Data Management : SQL, Power BI, MS Excel, Numpy, Pandas, DBMS

Software : Jupyter (Python), VS Code, GitHub, R Studio, OpenLCA, LINGO/LINDO, MS Excel

Soft Skills : Teamwork, Problem Solving, Leadership, Communication

PROJECTS

Mtech Thesis - Dynamic routing for effective Supply Chain Optimization of Solid Waste Management with IOT enabled bins. (June 2025 - Present)

- Designing a multi-objective model integrating IoT bin data for dynamic waste routing and facility allocation.
- Optimizing economic, environmental, and social sustainability.
- Applying real-time triggers and penalty logic to improve service and reduce emissions.

Portfolio website

(Nov - Dec, 2024)

Technologies Used: HTML, CSS, JavaScript

- Developing using HTML, CSS, and JavaScript to showcase my skills, projects, and background in a clean, responsive layout.

Project - Sustainable Concrete Innovation: Integrating Recycled Plastic for Enhanced Performance

(Oct - Nov, 2024)

Technologies: OpenLCA Software

- Analyzed recycled plastic (PET) as a replacement for cement and aggregates in concrete to enhance sustainability and optimize material properties.
- Assessed environmental, economic, and structural impacts using lifecycle analysis (LCA).
- Proposed sustainable solutions for concrete construction with improved thermal resistance and durability.

Project - Optimizing Sustainable Concrete Mixes with Recycled Plastic

(Sep - Nov, 2024)

Technologies Used: Multi-Objective Optimization (Genetic Algorithm), Simulation and Computational Tools

- Optimized concrete mix with recycled PET using genetic algorithms to reduce cost and CO₂ emissions.
- Designed sustainable mixes for single-story building components, meeting all constraints.

Carbon Capture and Modelling Approaches

(Nov 2024)

- Explored key carbon capture methods: pre-combustion, post-combustion, oxy-fuel, and direct air capture.
- Analyzed modelling techniques including CFD, dynamic, AI/ML, and hybrid models for process optimization.
- Assessed environmental and economic impacts to support sustainable carbon capture system design.

Life Cycle Assessment of Concrete Production

(Nov 2024)

Technology used - OpenLCA

- Performed cradle-to-gate LCA comparing traditional cement with SCM-based cement (calcined clay & limestone) using TRACI v2.1.
- Evaluated impacts across climate change, toxicity, and air quality, showing that clinker replacement significantly reduces environmental footprint.

SWOT Analysis of AI in Autonomous Vehicles

(March 2025)

- Analyzed strengths, weaknesses, opportunities, and threats of AI integration in self-driving cars.
- Evaluated key factors including safety, cost, adaptability, ethics, cybersecurity, and public acceptance.
- Highlighted AI's potential in smart city integration, shared mobility, and environmental sustainability.

Sustainable Bioethanol Supply Chain Design

(Feb 2025)

- Developed an MILP-based model to optimize bioethanol supply chain using second-generation biomass.
- Conducted techno-economic-environmental analysis including cost, GHGe, and carbon credit impacts.
- Assessed blending, transport, and import strategies to meet India's 20% ethanol mandate efficiently.

B.Tech Final Year Project - EV Battery Management System

- Designed a system to monitor and optimize EV battery performance, safety, and charging efficiency.
- Focused on managing voltage, temperature, and energy to enhance battery lifespan and prevent hazards.
- Enabled intelligent decision-making for charge/discharge cycles to improve energy utilization and system efficiency.

(Aug 2023 - May 2024)

Real-Time Emotion Detection Using OpenCV

Technologies Used: Python, OpenCV, Keras, NumPy, Haar Cascade, CNN

- Built a real-time system to detect human emotions from facial expressions using a pre-trained CNN model.
- Integrated OpenCV for face detection (Haar Cascade) and Keras for emotion classification into 7 categories (e.g., happy, sad, surprise).
- Applied image preprocessing, feature extraction, and prediction with live webcam feed.

(May - June, 2025)

Portfolio website

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(Nov - Dec, 2024)

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(Nov - Dec, 2024)

RELEVANT COURSES

Computer Science

- Operating System, Computer Network, SQL, DBMS (Youtube)

Data Analysis and Optimization

- Analysis and Methods for Sustainable Engineering, Supply Chain Management & Circular Economy, SWOT Analysis and Risk Management.

Sustainability

- Concepts in Sustainable Engineering, Sustainable Energy Technology: Energy Sources, Energy Efficiency, Storage and Optimization, Green Chemistry and Industrial Processes, Chemistry for Sustainable Energy.

Carbon Capture, Utilization and Storage

INTERNSHIP

N.E.Railway (Signal & Telecommunication)

- Research in Railway Signaling principles, Interlocking Principles and Railway System Of Working.
- Done research about the working of BPAC using UFSBI in railway.

(June 2023-July 2023)

POSITIONS OF RESPONSIBILITY

Final Year Project Team Leader : B.Tech

Aug 2023 - May 2024

Captain : MMMUT University Football Team

Aug 2022 - May 2024

Teaching Assistant : Sustainable Engineering, IIT Hyderabad

June 2025 - Present

ACHIEVEMENTS AND CERTIFICATIONS

District Level Kabaddi Player

Participated in Flipkart Grid 4.0

Qualified the 1 st round of Codekaze of Coding Ninjas

Completed Web Development Bootcamp with SHAPEAI

Completed workshop on Artificial Intelligence with IEEE, MMMUT

EXTRACURRICULAR

NCC B & C Certificate Holder

June 2020 – May 2023

Sports : Football, Volleyball, Chess